Software Engineering

Lecture One

2026 • Najwan abed

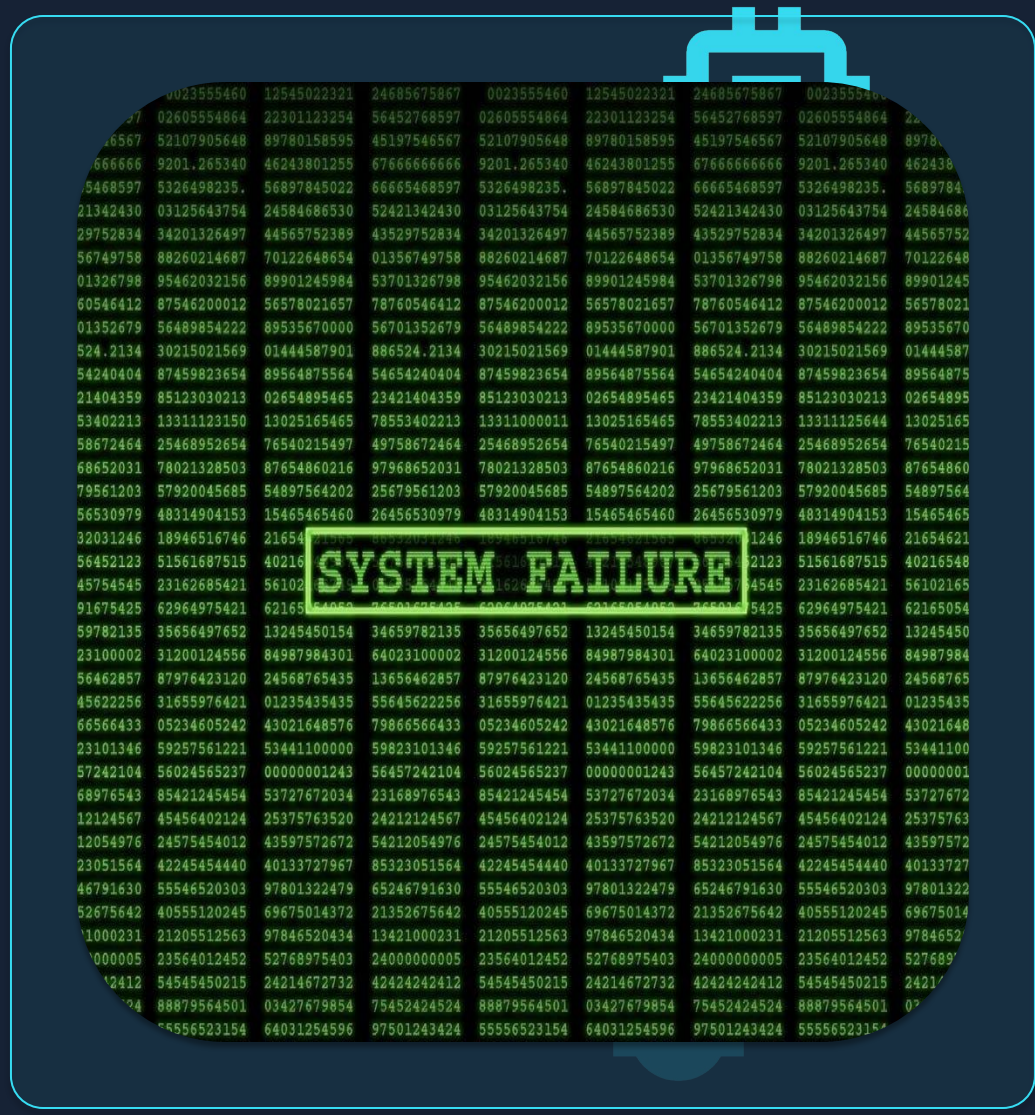
Software Engineering syllabus

- The ‘software Crisis’
- What is software
- Software Characteristics
- Software Applications
- Definitions of software engineering
- Software engineering layers
- The software process
- Software process models
- The linear sequential model
- The prototyping model
- The rad model
- Evolutionary software process models
- The management spectrum
- Testing
- Control graph

The Software Crisis

Software lags far behind the hardware's potential!

> Many large projects fail in that they're late, over budget, don't work well, or are abandoned



How Crisis Manifested

Over-budget

Projects running over-budget

Over-time

Projects running over-time

Inefficient

Software was very inefficient

Low Quality

Software was of low quality

Unmet Requirements

Software often did not meet requirements

Unmanageable

Projects were unmanageable and code difficult to maintain

Famous Software Failures

Therac 25

Radiation therapy machine failure

Cost lives

Arianne 5

Self-destructed 37 seconds after launch

$370 million losses

Y2K Bug

Millennium date rollover issue

Expensive scare

Pentium FDIV Bug

Floating point division error

$ Expensive scare

London Ambulance Service (LAS) 1992

Thoroughly documented project failure

London Ambulance Service 1992

2,000-3,000

Calls per day

~66%

Emergency (999) calls

200

Staff (of 2,700 total)

Paper Tickets

Details written on paper tickets for each call

Street Atlas

Map reference lookup in street atlas

Conveyor Belt

Tickets passed to central point

Central Point

Controller de-duplicates tickets

Average Response Time

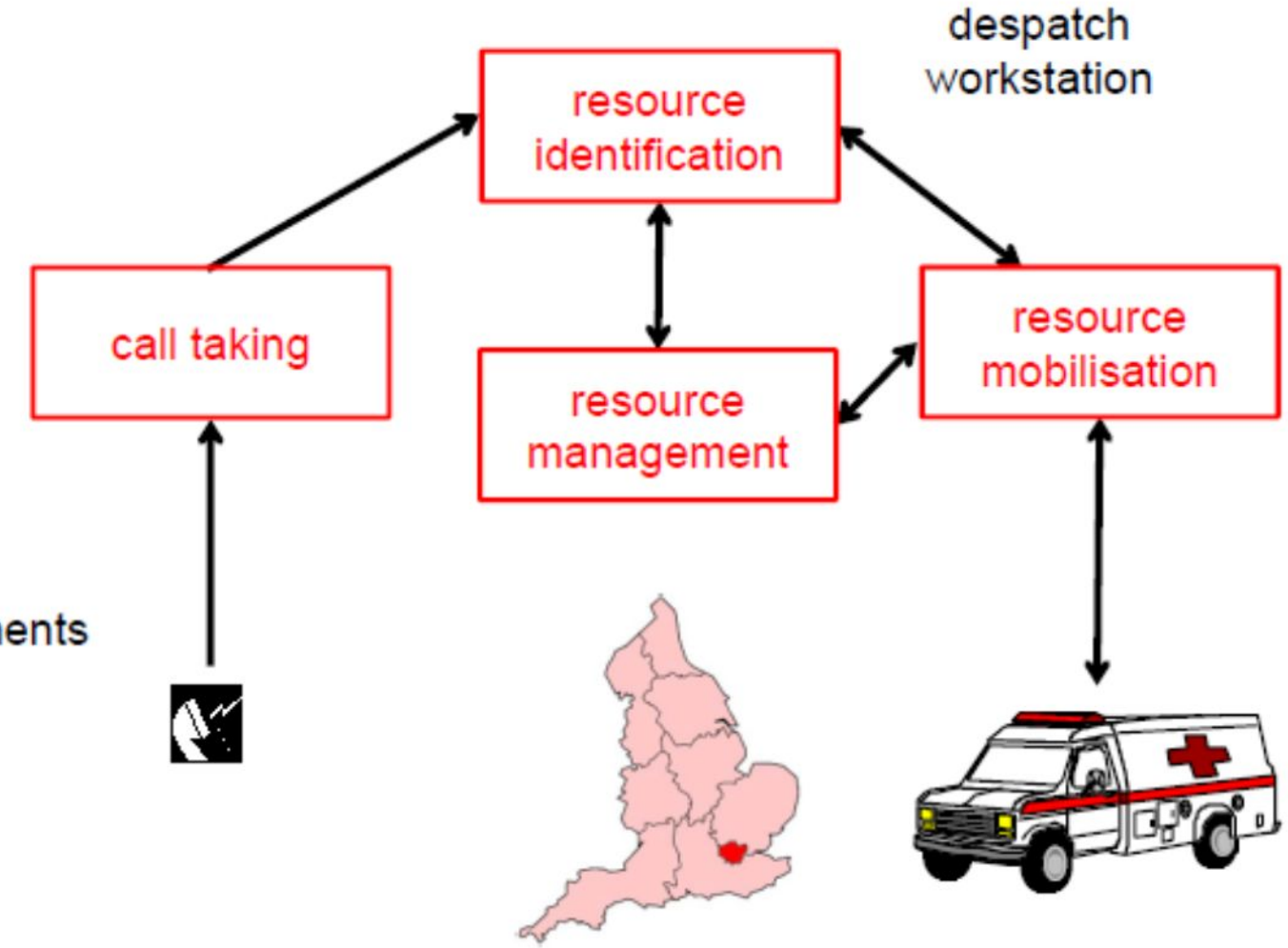
From call to dispatch

3 min

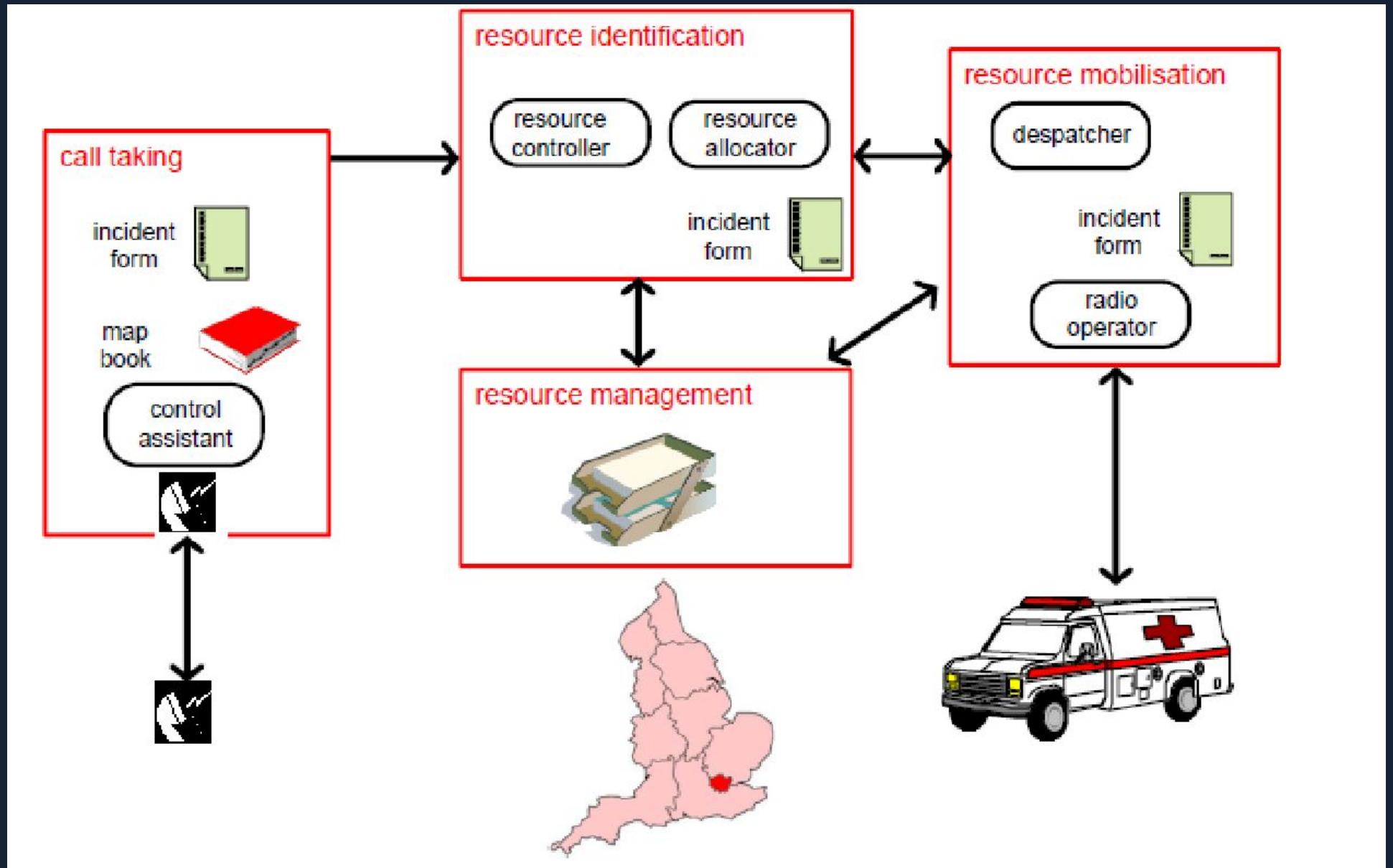
The manual dispatch system

PROPERTIES:

- Large
- Real-time
- Critical
- Data rich
- Embedded
- Distributed
- Mobile components



The manual implementation



Automation Attempt

1980s Failure

Automation attempt failed load test

Awarded To

Systems Options Ltd

Consortium with Apricot & Datatrak

Timeline Issue

Most firms said timeline was unrealistic

Tender Process

- Feb 1991: Tender sent
- Jan 1992: Deadline

Market Interest

35
Firms Looked

19
Proposed

Contract Price

£937K

£700K cheaper than next bidder

Project Failures

Ignored Advice

On cost and timescale

No Systems View

Lack of holistic approach

Inflexible Spec

Incomplete specification

Ignored Practices

Established work practices

Unrealistic Timeline

Timescale was unrealistic

Technical Fix

Change through technical system

> "They made virtually every mistake in the book"
>
> Paul Williams
> LAS Inquiry Member

Common mistakes in software project failures

Understanding Software

Software Characteristics

Intangible

Not physical, exists in digital form

Logic-Based

Comprises algorithms and data structures

Developed Not Manufactured

Created by engineers, not mass-produced

Software Applications

System Software

Operating systems, utilities

Application Software

Business, productivity tools

Engineering Software

CAD/CAM, simulation tools

Embedded Software

Real-time control systems

AI Software

Machine learning, neural networks

Web-Based Software

Cloud services, web apps

Software Engineering

Definition

The application of engineering principles to software development

Software Engineering Layers

Quality Layer

QA focus, testing, verification

Tools Layer

Automated support, CASE tools

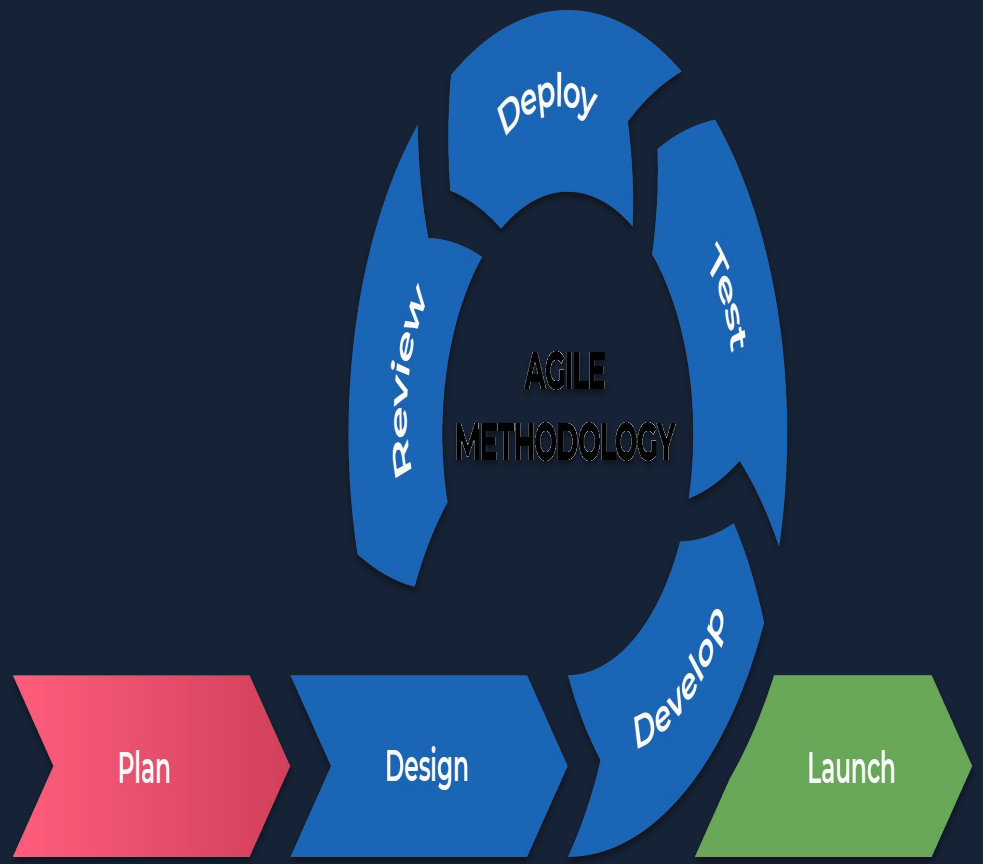
Methods Layer

How to do it better, techniques

Process Layer

How to do it, workflow models

Software Process Models



Linear Sequential Model

Waterfall approach, sequential phases

Prototyping Model

Rapid iteration, user feedback loop

RAD Model

Rapid Application Development

Evolutionary Models

Incremental, spiral approaches

The Management Spectrum

Process Management

Planning, scheduling, tracking

People Management

Team coordination, motivation

Technology Management

Methods, tools, automation

Quality Management

Testing, QA, standards

Testing

Ensuring software meets quality standards and requirements

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing

References & Further Reading

Primary Textbook

Software Engineering Ninth Edition

by Ian Sommerville

Key Topics Covered

- Software Crisis
- Software Characteristics
- Process Models
- Testin g
- Control Graph